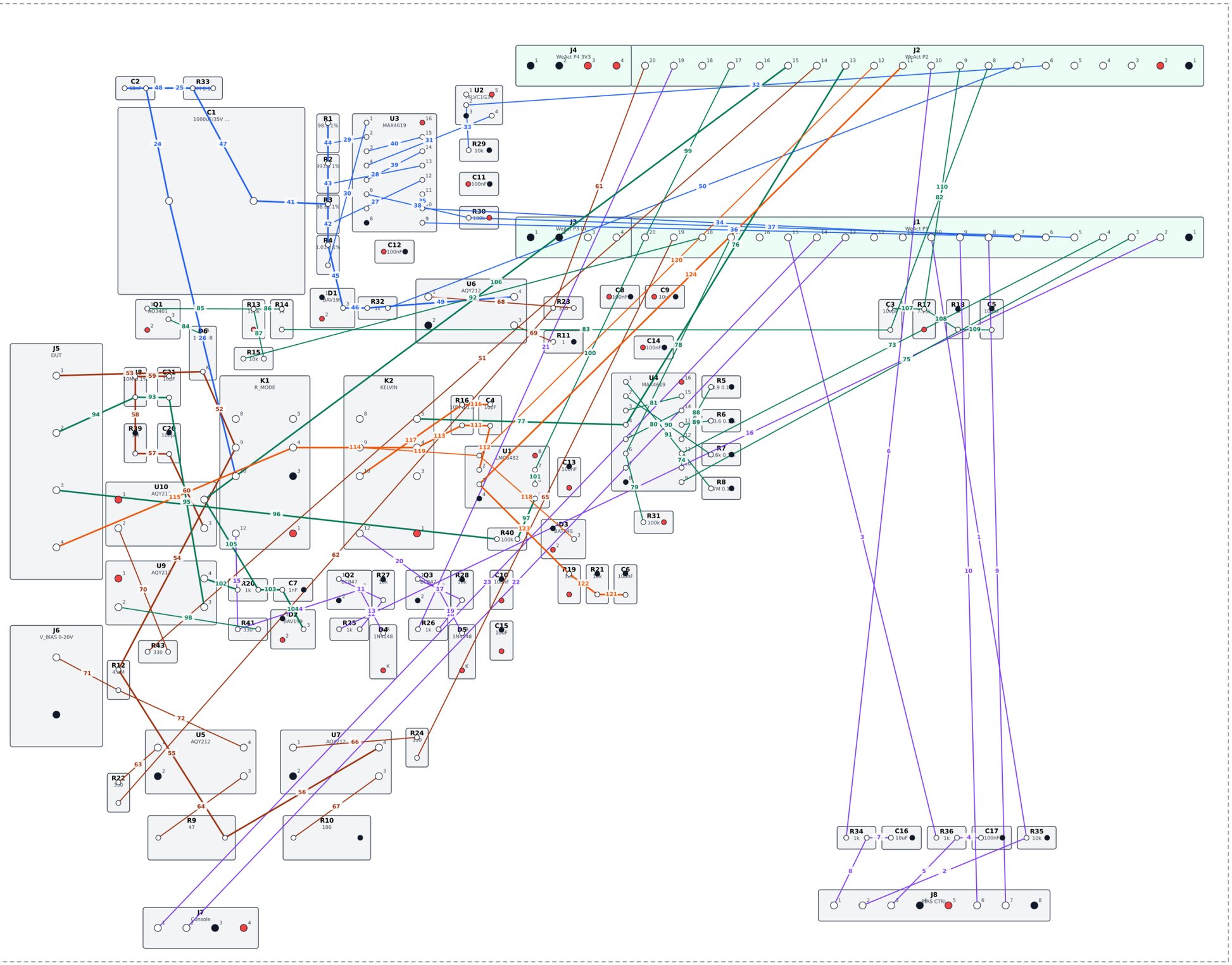
ESWCap Rev C.1 — wiring map (all blocks)

Parts are placed as on the PCB (top view). Each numbered line is one wire (ESWCap-wire-list.csv). Black pins → GND rail, red pins → +3V3 rail.



Legend

Power, relays & control wires

C-mode wires

Bias & safety wires

R-mode Kelvin wires

R-mode TIA wires

thick line = sensitive analog: keep short

● black pin = connect to GND rail

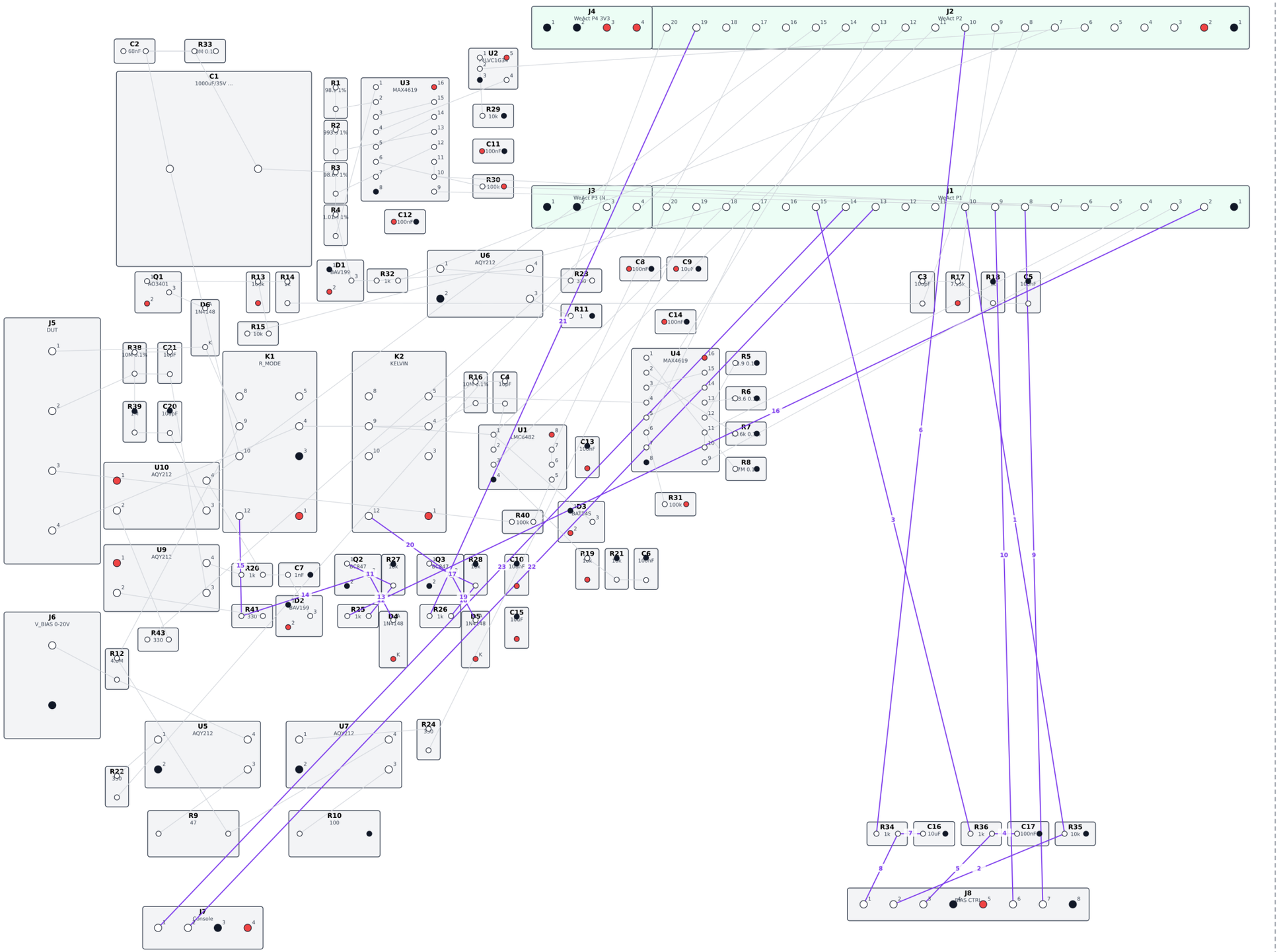
● red pin = connect to +3V3 rail

numbers on pins = pin numbers (K/A = diode)

wire numbers = ESWCap-wire-list.csv

ESWCap Rev C.1 — wiring map: Power, relays & control

Parts are placed as on the PCB (top view). Each numbered line is one wire (ESWCap-wire-list.csv). Black pins → GND rail, red pins → +3V3 rail.



Legend

Power, relays & control wires

C-mode wires

Bias & safety wires

R-mode Kelvin wires

R-mode TIA wires

thick line = sensitive analog: keep short

● black pin = connect to GND rail

● red pin = connect to +3V3 rail

numbers on pins = pin numbers (K/A = diode)

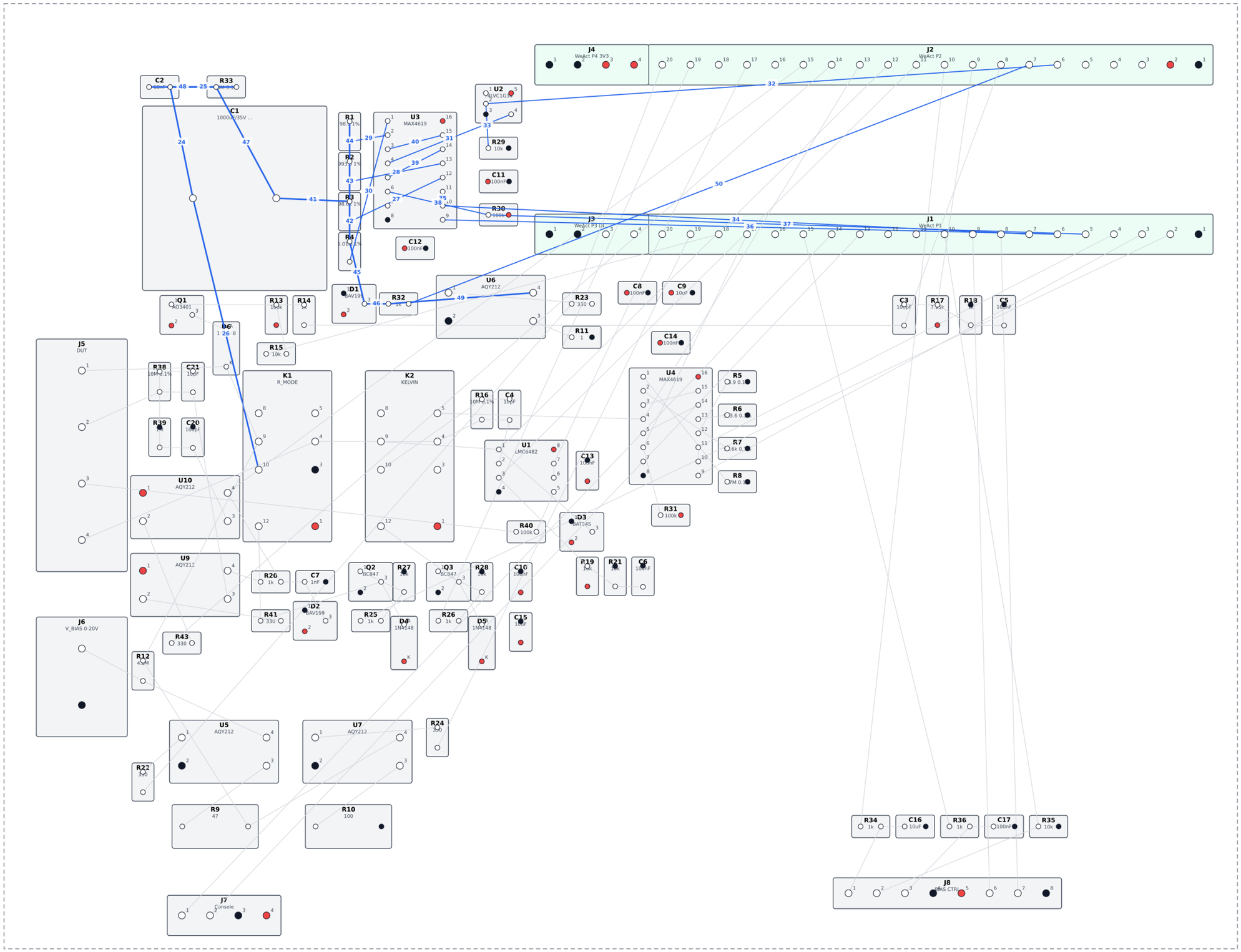
wire numbers = ESWCap-wire-list.csv

Wires on this page (Power, relays & control):

1 J1-10 → R35-1 (BIAS_EN)
2 R35-1 → J8-2 (BIAS_EN)
3 J1-15 → R36-1 (BIAS_MON)
4 C17-1 → R36-2 (BIAS_MON_F)
5 R36-2 → J8-3 (BIAS_MON_F)
6 J2-10 → R34-1 (BIAS_SET)
7 C16-1 → R34-2 (BIAS_SET_F)
8 R34-2 → J8-1 (BIAS_SET_F)
9 J1-8 → J8-7 (BIAS_UART_RX)
10 J1-9 → J8-6 (BIAS_UART_TX)
11 Q2-1 → R27-1 (K1_BASE)
12 R27-1 → R25-2 (K1_BASE)
13 D4-2 → Q2-3 (K1_C)
14 Q2-3 → R41-1 (K1_C)
15 R41-1 → K1-12 (K1_C)
16 J1-2 → R25-1 (K1_DRV)
17 Q3-1 → R28-1 (K2_BASE)
18 R28-1 → R26-2 (K2_BASE)
19 D5-2 → Q3-3 (K2_C)
20 Q3-3 → K2-12 (K2_C)
21 J2-19 → R26-1 (K2_DRV)
22 J1-13 → J7-2 (UART_RX)
23 J1-14 → J7-1 (UART_TX)

ESWCap Rev C.1 — wiring map: C-mode

Parts are placed as on the PCB (top view). Each numbered line is one wire (ESWCap-wire-list.csv). Black pins → GND rail, red pins → +3V3 rail.



Legend

Power, relays & control wires

C-mode wires

Bias & safety wires

R-mode Kelvin wires

R-mode TIA wires

thick line = sensitive analog: keep short

● black pin = connect to GND rail

● red pin = connect to +3V3 rail

numbers on pins = pin numbers (K/A = diode)

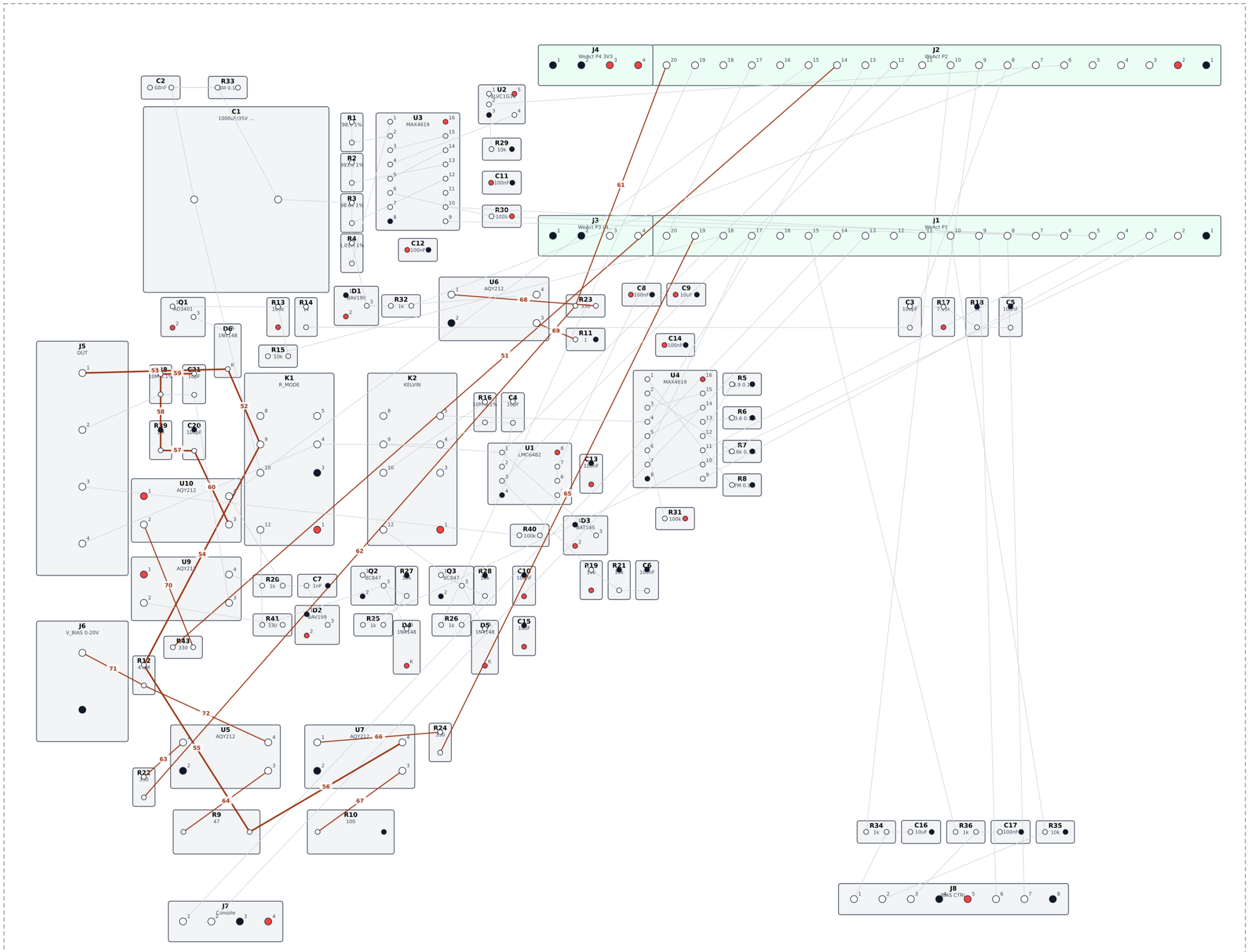
wire numbers = ESWCap-wire-list.csv

Wires on this page (C-mode):

24 C1-2 → C2-2 (C_BLOCK) ⚠
25 C2-2 → R33-2 (C_BLOCK) ⚠
26 C1-2 → K1-10 (C_BLOCK) ⚠
27 R3-1 → U3-12 (C_CH0)
28 R2-1 → U3-13 (C_CH1)
29 R1-1 → U3-2 (C_CH2)
30 R4-1 → U3-1 (C_CH3)
31 U2-4 → U3-4 (C_MUX_COM)
32 J2-6 → U2-2 (DRIVE)
33 U2-2 → R29-1 (DRIVE)
34 J1-7 → U3-10 (MUXC_A0)
35 U3-10 → U3-11 (MUXC_A0)
36 J1-6 → U3-9 (MUXC_A1)
37 J1-5 → R30-1 (MUXC_EN)
38 R30-1 → U3-6 (MUXC_EN)
39 U3-14 → U3-5 (MUXC_P)
40 U3-15 → U3-3 (MUXC_Q)
41 C1-1 → R3-2 (VCAP) ⚠
42 R3-2 → R4-2 (VCAP) ⚠
43 R3-2 → R2-2 (VCAP) ⚠
44 R2-2 → R1-2 (VCAP) ⚠
45 R4-2 → D1-3 (VCAP) ⚠
46 D1-3 → R32-1 (VCAP) ⚠
47 C1-1 → R33-1 (VCAP) ⚠
48 R33-1 → C2-1 (VCAP) ⚠
49 R32-1 → U6-4 (VCAP) ⚠
50 J2-7 → R32-2 (VCAP_ADC)

ESWCap Rev C.1 — wiring map: Bias & safety

Parts are placed as on the PCB (top view). Each numbered line is one wire (ESWCap-wire-list.csv). Black pins → GND rail, red pins → +3V3 rail.



Legend

Power, relays & control wires

C-mode wires

Bias & safety wires

R-mode Kelvin wires

R-mode TIA wires

thick line = sensitive analog: keep short

● black pin = connect to GND rail

● red pin = connect to +3V3 rail

numbers on pins = pin numbers (K/A = diode)

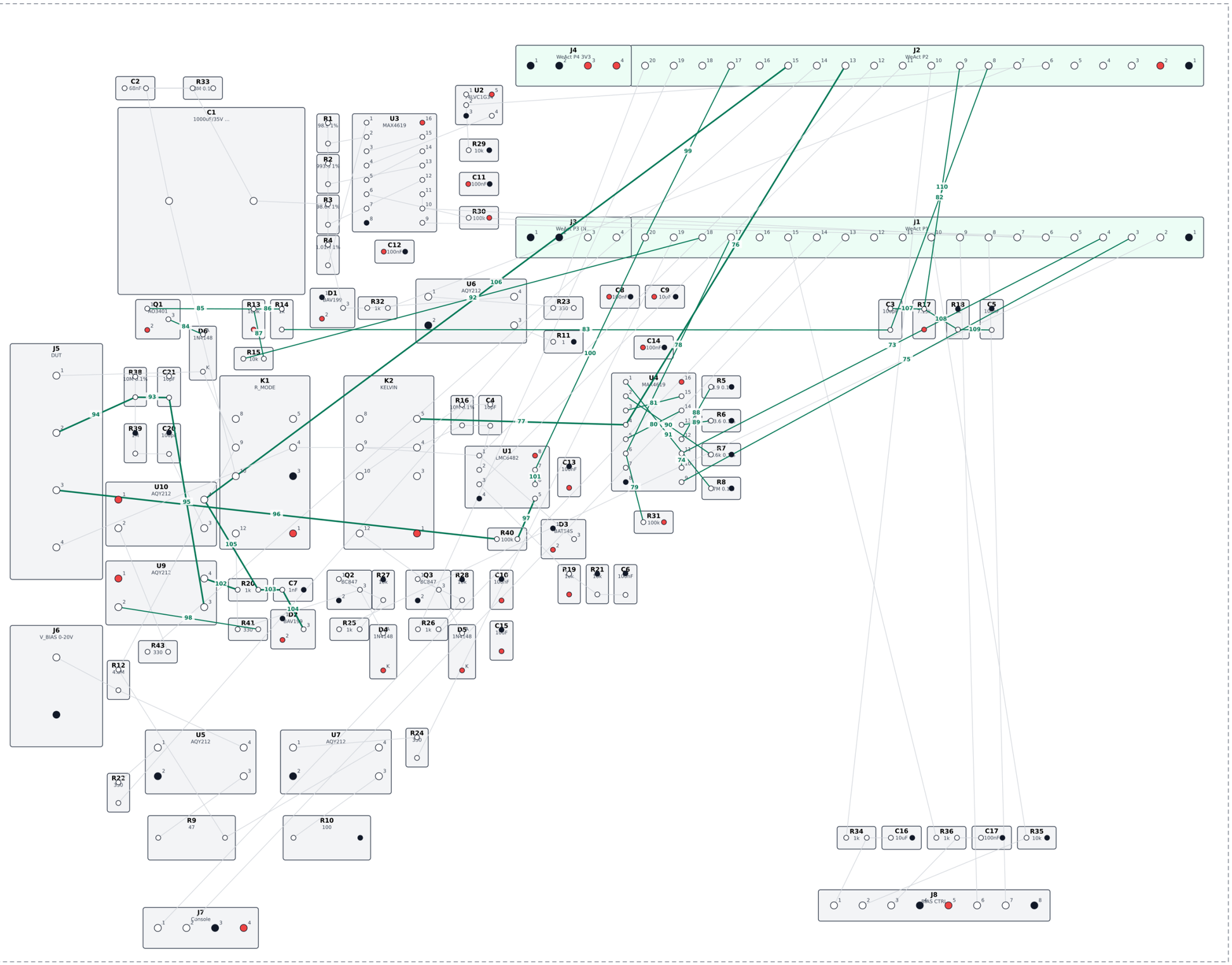
wire numbers = ESWCap-wire-list.csv

Wires on this page (Bias & safety):

51 J2-14 → R43-1 (C_MODE_SW)
52 D6-1 → K1-9 (FH) ⚠
53 D6-1 → J5-1 (FH) ⚠
54 K1-9 → R12-2 (FH) ⚠
55 R12-2 → R9-2 (FH) ⚠
56 R9-2 → U7-4 (FH) ⚠
57 C20-1 → R39-1 (SH_DIV) ⚠
58 R39-1 → R38-2 (SH_DIV) ⚠
59 R38-2 → C21-2 (SH_DIV) ⚠
60 C20-1 → U10-3 (SH_DIV) ⚠
61 J2-20 → R23-1 (SSR13)
62 R23-1 → R22-1 (SSR13)
63 R22-2 → U5-1 (SSR1_LED)
64 R9-1 → U5-3 (SSR1_SW)
65 J1-19 → R24-1 (SSR2_CTRL)
66 R24-2 → U7-1 (SSR2_LED)
67 R10-1 → U7-3 (SSR2_SW)
68 R23-2 → U6-1 (SSR3_LED)
69 R11-1 → U6-3 (SSR3_SW)
70 R43-2 → U10-2 (U10_LED_K)
71 J6-1 → R12-1 (V_BIAS)
72 R12-1 → U5-4 (V_BIAS)

ESWCap Rev C.1 — wiring map: R-mode Kelvin

Parts are placed as on the PCB (top view). Each numbered line is one wire (ESWCap-wire-list.csv). Black pins → GND rail, red pins → +3V3 rail.



Legend

Power, relays & control wires

C-mode wires

Bias & safety wires

R-mode Kelvin wires

R-mode TIA wires

thick line = sensitive analog: keep short

● black pin = connect to GND rail

● red pin = connect to +3V3 rail

numbers on pins = pin numbers (K/A = diode)

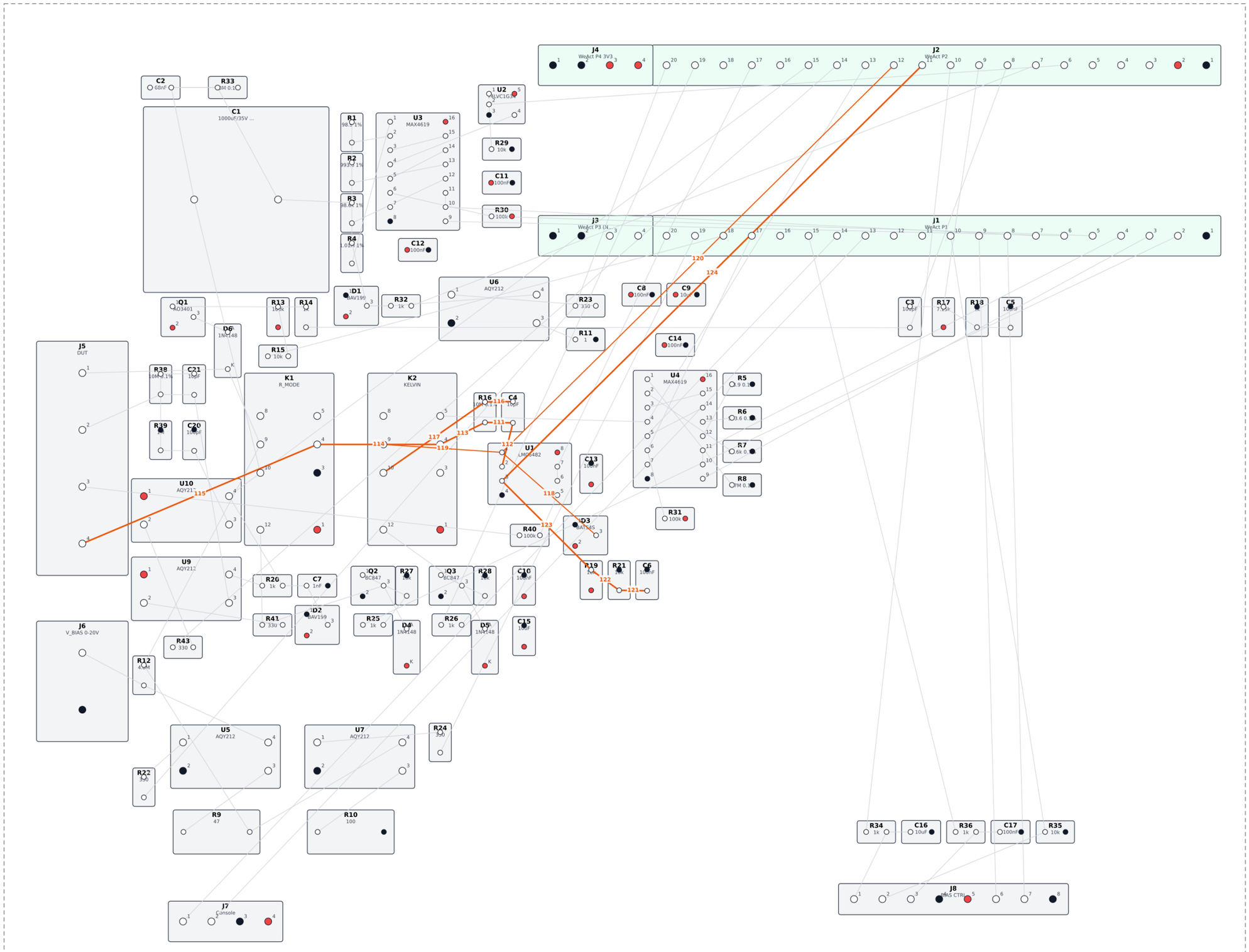
wire numbers = ESWCap-wire-list.csv

Wires on this page (R-mode Kelvin):

73 J1-4 → U4-11 (MUXR_A0)
74 U4-11 → U4-10 (MUXR_A0)
75 J1-3 → U4-9 (MUXR_A1)
76 J2-13 → U4-4 (MUXR_COM) ⚠
77 U4-4 → K2-5 (MUXR_COM) ⚠
78 J1-17 → U4-6 (MUXR_EN)
79 U4-6 → R31-1 (MUXR_EN)
80 U4-14 → U4-5 (MUXR_P)
81 U4-15 → U4-3 (MUXR_Q)
82 C3-1 → J2-8 (OP1_OUT)
83 C3-1 → R14-1 (OP1_OUT)
84 D6-2 → Q1-3 (Q1_D)
85 Q1-1 → R13-2 (Q1_GATE)
86 R13-2 → R14-2 (Q1_GATE)
87 R13-2 → R15-2 (Q1_GATE)
88 R5-1 → U4-12 (R_CH0)
89 R6-1 → U4-13 (R_CH1)
90 R7-1 → U4-2 (R_CH2)
91 R8-1 → U4-1 (R_CH3)
92 J1-18 → R15-1 (R_FORCE)
93 C21-1 → R38-1 (SH) ⚠
94 R38-1 → J5-2 (SH) ⚠
95 C21-1 → U9-3 (SH) ⚠
96 J5-3 → R40-1 (SL) ⚠
97 R40-2 → U1-5 (SL_BUF) ⚠
98 R41-2 → U9-2 (U9_LED_K)
99 J1-20 → J2-17 (VBUF)
100 J1-20 → U1-7 (VBUF)
101 U1-7 → U1-6 (VBUF)
102 R20-1 → U9-4 (VN_IN) ⚠
103 C7-1 → R20-2 (VN_SENSE) ⚠
104 C7-1 → D2-3 (VN_SENSE) ⚠
105 R20-2 → U10-4 (VN_SENSE) ⚠
106 U10-4 → J2-15 (VN_SENSE) ⚠
107 C3-2 → R17-2 (VREF)
108 R17-2 → R18-1 (VREF)
109 R18-1 → C5-1 (VREF)
110 R17-2 → J2-9 (VREF)

ESWCap Rev C.1 — wiring map: R-mode TIA

Parts are placed as on the PCB (top view). Each numbered line is one wire (ESWCap-wire-list.csv). Black pins → GND rail, red pins → +3V3 rail.



Legend

Power, relays & control wires

C-mode wires

Bias & safety wires

R-mode Kelvin wires

R-mode TIA wires

thick line = sensitive analog: keep short

● black pin = connect to GND rail

● red pin = connect to +3V3 rail

numbers on pins = pin numbers (K/A = diode)

wire numbers = ESWCap-wire-list.csv

Wires on this page (R-mode TIA):

111 C4-1 → R16-1 (FL) ⚠
112 C4-1 → U1-2 (FL) ⚠
113 R16-1 → K2-4 (FL) ⚠
114 K2-4 → K1-4 (FL) ⚠
115 K1-4 → J5-4 (FL) ⚠
116 C4-2 → R16-2 (RF_TOP) ⚠
117 R16-2 → K2-10 (RF_TOP) ⚠
118 D3-3 → U1-1 (TIA_OUT)
119 U1-1 → K2-9 (TIA_OUT)
120 U1-1 → J2-12 (TIA_OUT)
121 C6-1 → R21-1 (VBIAS) ⚠
122 R21-1 → R19-2 (VBIAS) ⚠
123 R19-2 → U1-3 (VBIAS) ⚠
124 U1-3 → J2-11 (VBIAS) ⚠